Abhinita Sanabada

ML and AI Engineer

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Seeking full-time ML engineering and AI engineering roles. Available to start now.

SUMMARY

ML and AI engineer who builds LLM agents, retrieval systems, guardrailed AI workflows, and on-device inference, backed by 10 years of production software engineering at Wells Fargo, JPMorgan Chase, and SAP Labs. M.S. Data Science candidate at San José State and AI engineering intern at a healthcare startup. Focused on AI systems that are evaluated, observable, and safe to run in production.

SKILLS

- **LLM / GenAI:** LangChain, LangGraph, RAG, embeddings and semantic search, MCP, tool calling, multi-agent workflows, NeMo Guardrails, human-in-the-loop design
- **Model serving and training:** vLLM, LoRA fine-tuning, PyTorch, scikit-learn, Spark ML, CNNs, BiLSTMs, attention, self-consistency sampling
- **Evaluation:** calibration and held-out test splits, Macro-F1, class-imbalance handling (focal loss, augmentation), error analysis, benchmarking
- **Data and infrastructure:** Python, SQL, Spark/PySpark, Kafka, Airflow, Snowflake, FastAPI, Docker, Kubernetes, AWS, Azure, Redis, MongoDB, PostgreSQL

AI AND ML PROJECTS

EdgeSupport: on-device IT triage agent demo video code *HP Edge AI Hackathon, SJSU, 2026*

- Built an IT help-desk agent that runs Qwen2.5-7B locally with vLLM on an HP ZGX Nano (NVIDIA GB10), so private device logs never leave the site.
- Designed self-consistency routing: 3 to 5 samples per ticket, kept on site only when answers agree and cite telemetry evidence; otherwise escalated with a reason code.
- Built the privacy-redaction and telemetry pipeline and led team integration (11 of 16 commits); reached 95% accuracy vs. 75% for a rules-only baseline on a held-out set, with zero PII leaked.

Agentic Recruiting Workflow Simulation code *Kafka, FastAPI, LangChain, Redis, AWS*

- Built agents for resume parsing, job matching, and outreach drafting with human approval, on Kafka pipelines with idempotent consumers; benchmarked on 10K records.

Speech Emotion Recognition with Dialogue Context code *PyTorch, CNN, BiLSTM, Attention*

- CNN, BiLSTM, and attention over Log-Mel and MFCC features with focal loss for class imbalance; 52.4% Macro-F1 and 55.9% accuracy on IEMOCAP, beating MLP and transformer baselines.

Healthcare Affordability Semantic Search code *PySpark, Spark ML, HDFS, MiniLM*

- Cleaned 13.6M+ rows of CMS pricing data into ~47K records with Spark; added MiniLM semantic search and Spark regression price estimates.

EXPERIENCE

Healthcare Startup (Confidential) | *AI Engineer Intern* Jun 2026 – Present

- Build multi-agent LangChain/LangGraph workflows for patient-data and clinical-document processing.
- Add NVIDIA NeMo Guardrails safety checks and structured outputs; build ingestion and feature-extraction for downstream ML.

Wells Fargo | *Senior Software Engineer* Nov 2021 – Jan 2024

- Prototyped an LLM and RAG ops copilot over runbooks, Splunk, and ServiceNow to help triage production breaks, with human approval.
- Owned Java/Spring services and SQL processing for futures trade, margin, and risk; tuned stored procedures for ~30% faster runtimes.

SAP Labs | *Software Development Engineer 2* Apr 2019 – Oct 2021

- Delivered an automotive orchestration solution on SAP HANA Cloud; owned production reliability and a Kubernetes POC.

JPMorgan Chase & Co. | *Software Engineer* Jan 2017 – Apr 2019

- Built Spring Boot services and Airflow/Spark ETL with data-quality checks for asset and wealth management.

Capgemini | *Software Engineer* Apr 2014 – Aug 2016

- Built banking reports in Core Java, Oracle SQL, and Power BI/Tableau.

EDUCATION

M.S. Data Science – San José State University Expected May 2027

B.Tech. Electronics and Communication Engineering – Biju Patnaik University of Technology 2008 – 2012

Certifications: Machine Learning Engineering (Andrew Ng), AWS Fundamentals, Microsoft Azure Fundamentals